



N O N - N O R M A T I V E

$T = \text{sigma_v} \quad \text{det} = -1 \quad (x, y) \rightarrow (-x, y)$

"This reflects the standard."

Informative / Advisory (NOTE)

Owl Semaphore — Non-Normative

OWL 2 / Reflection State (sigma_v) / Standard Specification

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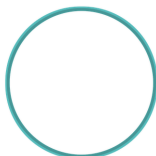
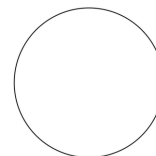
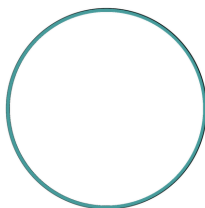
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CONCEPT-DOI 10.5281/zenodo.19473697 PREVIOUS-VERSION-DOI 10.5281/zenodo.20433053 (v2.0.2)
PRIOR-V2.0.1-DOI 10.5281/zenodo.20419874 PRIOR-V2.0.0-DOI 10.5281/zenodo.20418539 PRIOR-V1.2.0-DOI 10.5281/zenodo.19474599
SOURCE github.com/IT-Help-San-Diego/owl-semaphore VERSION v3.0.0 · LICENSE CC-BY-4.0

BANNER-TUPLE :: STATE=NON-NORMATIVE :: TRANSFORM= $T = \text{sigma_v} \quad \text{det} = -1 \quad (x, y) \rightarrow (-x, y)$:: QUOTE="This reflects the standard." :: VERSION=v3.0.0 ::
VERSION-DOI=10.5281/zenodo.19473697 :: CONCEPT-DOI=10.5281/zenodo.19473697 :: PREVIOUS-VERSION-DOI=10.5281/zenodo.20433053

NON-NORMATIVE — LAYER PROOF PALETTE

**OWL-2 NON-NORMATIVE approved master asset kit — layer proof**

Selected visual: Math Mirror Center Scale 97 + Seam 17 + Five Over. Layers below compose to the final composite.

L0-inner-field-underpaint-17
bbox=(149, 149, 932, 932)L1-inner-teal-ring-outward-17
bbox=(134, 134, 947, 947)L2-meander-ring-original
bbox=(26, 26, 1055, 1055)L2.5-inner-meander-black-edge-5-over
bbox=(137, 137, 943, 943)L3-owl-math-mirror-center-scale-97
bbox=(226, 226, 841, 841)L4-outer-teal-ring
bbox=(4, 4, 1077, 1077)FINAL-COMPOSITE
bbox=(4, 4, 1077, 1077)

1. Statement of Intent

This document defines the **NON-NORMATIVE owl** as a formal epistemic state within the Owl Semaphore system.

This is not a decorative variation of the normative mark. It is a mathematically defined reflection state with a specific role in analysis, interpretation, and structured disagreement.

The goal is to preserve rigor while allowing controlled deviation from the baseline framework.

1A. The Story Before the Math — *Da Vinci's Wings*

$$\mathbf{T} = \sigma_v \cdot \det = -1 \cdot (x, y) \rightarrow (-x, y)$$

Stand in front of a mirror. You are still upright — up is still up, down is still down — but left and right have swapped. That is a vertical-axis reflection: $(x, y) \rightarrow (-x, y)$. Not a rotation — a mirror image.

When Leonardo da Vinci spent years studying birds and sketching wing mechanics — preserved in the *Codex on the Flight of Birds* and related manuscripts now held by the Biblioteca Reale di Torino (Library of Congress, *Codex on the Flight of Birds*; Smithsonian Air and Space, “Leonardo da Vinci’s Codex on the Flight of Birds”) — he was facing the other direction from the prevailing



assumption that powered human flight was beyond reach. His machines did not become practical aircraft. What he left behind was a rigorous exploratory record: anatomical observation, geometric reasoning, and structured speculation.

The first powered, sustained, controlled heavier-than-air flight is attributed to the Wright Brothers at Kitty Hawk on 17 December 1903 (Library of Congress, Wright Brothers collection; Smithsonian National Air and Space Museum, *1903 Wright Flyer*). That success arrived four centuries after Leonardo, inside an accumulated aeronautical history of gliders, propulsion experiments, and aerodynamic theory — not as a direct line from his notebooks, but inside the broader practice of non-normative work that had matured into something actionable. The historical claim made here is the modest one: structured, rigorous exploration that has not yet succeeded is still part of the engine of progress.

The NON-NORMATIVE owl marks that posture — the mirror state, facing what the canonical view has its back to, **without claiming to have replaced the canonical view yet**. It is reflection, not rejection. Non-normative work is the engine of progress: rigorous exploration that has not finished yet.

Bridge — from the story to the operator. The mirror in *Da Vinci's Wings* is a precise statement of the formal mapping, not a metaphor laid over it. Standing upright while left and right swap is exactly the vertical-axis reflection $\sigma_v: (x, y) \rightarrow (-x, y)$, determinant -1 (§4). The “still upright” part matters — σ_v *preserves the vertical reference* (up stays up) while reversing the lateral stance; that is why NON-NORMATIVE is an *alternative orientation* to the baseline, not its negation or its inversion (the full inversion is CRITICAL's C_2). Leonardo facing the direction the prevailing view had its back to — rigorous, structured, not yet successful — *is* what the reflected determinant encodes: **legitimate exploration, not failure and not error**. The $\det = -1$ marks orientation reversal; it does not mark wrongness. Read the story as the intuition and §4 as its formalization; they are the same claim at two resolutions.

This story is the human-intuition bridge to the mathematical formalism in §§2 onward. The deliberate ordering is story $\rightarrow$ transform $\rightarrow$ scientific use $\rightarrow$ objections/verification, so a reader who would argue the σ_v state from intuition, from history-of-science, or from the V_4 algebra can each enter through the right door.

2. System Context

The Owl Semaphore system is defined by the Klein four-group:

$$V_4 = \{I, \sigma_v, C_2, \sigma_h\}$$

The NON-NORMATIVE owl corresponds to the vertical reflection operator:

$$\sigma_v : (x, y) \mapsto (-x, y)$$



3. Ontological Role

3.1 Semantic Designation

NON-NORMATIVE represents:

- legitimate alternative interpretation
- reflective critique
- analytical deviation
- structured disagreement

3.2 Interpretive Role

This state indicates that the content:

- departs from the canonical model
- remains structurally grounded in the same system
- is not arbitrary or invalid

3.3 What It Does Not Mean

- not random
- not incorrect by default
- not adversarial (that is CRITICAL)

It is **reflection**, not rejection.

4. Mathematical Definition

4.1 State Operator

$$T_{\text{non}} = \sigma_v$$

4.2 Matrix Form

$$\sigma_v = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

4.3 Determinant

$$\det(\sigma_v) = -1$$

4.4 Properties



- orientation-reversing
- reflection class
- order 2

$$\sigma_v^2 = I$$

5. Coordinate System

The NON-NORMATIVE state uses the same coordinate system as NORMATIVE:

- canvas: 1080×1080
- center: (540, 540)

Transformation is applied relative to this center.

6. Canonical Orientation

6.1 Visual Definition

- upright
- faces LEFT

6.2 Transform Relationship

The NON-NORMATIVE owl is the horizontal mirror of the normative owl.

7. Asset Topology

Layer structure is identical to normative:

- L1 — inner field
- L2 — meander ring
- L3 — owl body
- L4 — outer ring

7.1 Composite Definition

$$N_{\text{non}} = L_1 \oplus L_2 \oplus L_3 \oplus L_4$$

8. Geometry



All geometric constraints are inherited from the normative standard:

- identical radii
- identical center
- identical annular structure

No geometric deformation is permitted.

9. Color Specification

9.1 Palette

- outer ring: #316964 (teal)
- owl: #316964 (teal)
- field: #d2d8d6 (cool gray)
- meander: unchanged (gold)

9.2 Color Doctrine

Teal represents analytical distance from the normative baseline while maintaining structural coherence.

10. Transparency and Alpha

Same rules as normative:

- RGBA required
 - corner alpha = 0
 - center alpha = 255
-

11. Provenance

11.1 Construction

Derived from normative by:

- horizontal reflection
- no rotation
- no scaling

11.2 Transform Integrity



The transform must be exact. Any distortion invalidates the state.

12. Asset Invariants

12.1 Algebraic

- operator = σ_v
- determinant = -1

12.2 Visual

- upright
- mirrored orientation

12.3 Structural

- geometry unchanged
 - layer structure unchanged
-

13. Integrity Regime

All assets must:

- pass SHA-3-512 verification
 - be reproducible from layers
-

14. Interpretation Rules

14.1 Positive Rule

When present:

The content reflects the normative framework while offering a structured alternative.

14.2 Negative Rule

It does not indicate failure or error.

15. Non-Permitted Changes



- rotation
- vertical flip
- C2 inversion
- geometry alteration

16. Relationship to Other States

- NORMATIVE → identity (“*This is the standard.*”)
- NON-NORMATIVE → reflection (“*This reflects the standard.*”)
- CRITICAL → inversion (“*This inverts the standard.*”)
- METACOGNITIVE → frame-audit (“*The observer audits the frame.*” — thinking examines its own frame)

17. Formal Definition

Let L_1 – L_4 be the layer fields.

$$N_{\text{non}} = L_1 \oplus L_2 \oplus L_3 \oplus L_4$$

with

$$T = \sigma_v$$

17A. Limitations and Scope

The NON-NORMATIVE state carries a specific misuse risk: it must never be read as a verdict of “wrong.” σ_v reverses orientation ($\det = -1$), not correctness — the state marks a *legitimate alternative facing*, and the whole reason it exists as a distinct, protected state is to keep structured exploration and dissent from collapsing into “not normative” and being dismissed (see the use-boundaries in §1C.1 of `OWL-SEMAPHORE-EXPLANATION.md`). Tagging an analysis NON-NORMATIVE to sideline it, rather than to describe the reflective move actually performed, is a misuse. Like the system as a whole, this is a coarse four-way stance classifier (see §12A of `OWL-SEMAPHORE-SYSTEM.md`): it does not grade how promising the exploration is, its epistemic reliability across annotators is not yet empirically validated, and applying the badge is itself a contestable evaluative act.

18. Closing Statement

The NON-NORMATIVE owl preserves system integrity while enabling structured deviation.

It is the mechanism by which the system permits disagreement without collapse.



OWL SEMAPHORE SYSTEM — CLASSIFICATION LEDGER (v3.0.0)



NORMATIVE

$T = I \quad \det = +1$
 $(x, y) \rightarrow (x, y)$
“This is the standard.”
RFC 2119 MUST / SHALL



NON-NORMATIVE

$T = \sigma_v \quad \det = -1$
 $(x, y) \rightarrow (-x, y)$
“This reflects the standard.”
Informative / Advisory (NOTE)



CRITICAL

$T = C_2 \quad \det = +1$
 $(x, y) \rightarrow (-x, -y)$
“This inverts the standard.”
RFC 2119 MUST NOT / SHALL NOT



METACOGNITIVE

$T = \sigma_h \quad \det = -1$
 $(x, y) \rightarrow (x, -y)$
“The observer audits the frame.”
Epistemic / Framework (META)

Accessibility: color is not the only carrier. State identity is recoverable from color + orientation + textual label (WCAG 2.2 SC 1.4.1).